

ECE 332 – ANTENNA LAB

INTRODUCTION

In this lab, you will build and measure the performance of a patch antenna. **Each student will make one and measure it, so each pair of lab partners will have two antennas and you will make use of both at the same time.** If instructions are followed and care taken in its construction, this antenna may work on a home device, like your wireless router.

For lab reports, please answer all questions and provide all plots/screen captures requested in the Procedure sections. One lab report per group member, groups can be no larger than two people, and include both names on the lab report. Questions are not labelled (life isn't a textbook!); you'll have to read this document carefully to answer all of them. The basis of your grade for this lab will be as follows:

- 50% for following all directions – plots set up exactly as described, correct process being followed
- 25% for full participation – you completed every step and didn't skip sections, you asked questions about sections you didn't understand, and the report was submitted on time.
- 25% for getting correct answers within an acceptable margin of error.

REQUIRED HARDWARE

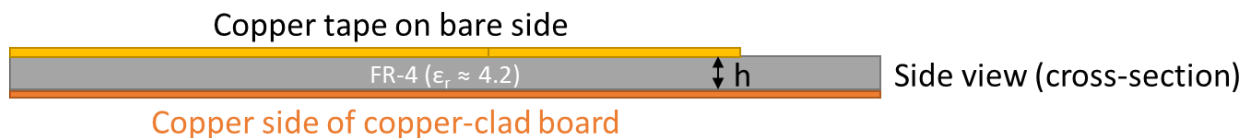
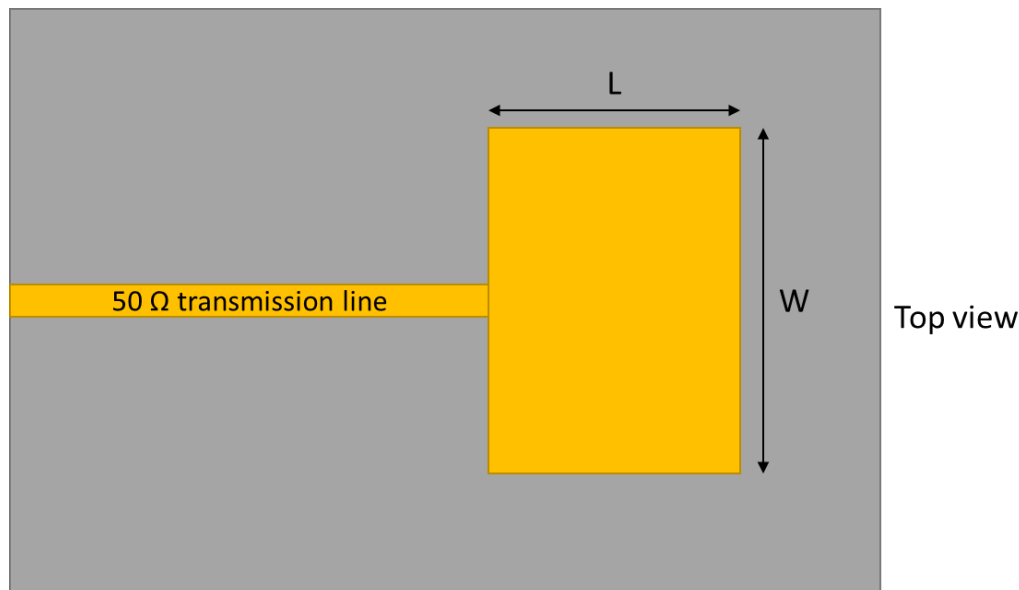
- Student Supplied: Lab Kit
 - Copper-clad FR4, single sided (i.e. only one side has copper on it)
 - Copper tape
 - Side-launch SMA connector (one per antenna)
- Lab Provided: VNA (located in the emag lab)
 - a NanoVNA is acceptable, but you'll get better results with the real thing.
 - Calibration Kit for VNA
 - If you're in the emag lab, you can use the eCal module (be gentle!), or you can use a mechanical standards kit. If you've never used the eCal module before, ask the TA.
 - If you don't know how to calibrate a VNA with the mechanical standards kit, ask the TA.
 - Soldering iron & peripherals
 - Xacto knife (you may have your own, too)

PROCEDURE

VNA SET-UP

1. Turn on the VNA if it's not on already. Don't calibrate it yet.
2. Connect the eCal module via USB cable to any of the VNAs in the room and let it warm up.
 - a. The light on the eCal module will turn green when it's ready to use.
 - b. You'll all be sharing the same eCal module, leave it plugged into USB when not in use, so it stays warmed up.
3. Format the VNA to display the data we want to measure
 - a. The 'Display' button will let you split the screen however you want. Set a 4-way quadrant split.
 - b. The 'Format' button lets you choose between Smith/Log Mag/VSWR/etc. formats
 - c. The 'Meas' button sets which S-parameter we want to measure
 - d. **On the top left quadrant of the screen, display a Log-Mag format measuring S_{11}**
 - e. **On the top right quadrant of the screen, display a Smith format ($R+jX$) measuring S_{11}**
 - f. **On the bottom left quadrant of the screen, display a Log-Mag format measuring S_{22}**
 - g. **On the bottom right quadrant of the screen, display a Smith format ($R+jX$) measuring S_{22}**
 - i. The VNAs have touch-screens. You may have to tap on a plot to select it before changing its settings.
 - h. **Set the Start frequency to 1.5 GHz and the Stop frequency to 2.7 GHz**
 - i. Connect an SMA cable (the blue ones from the emag lab) to each of the two ports on the VNA.
 - i. the cables from your lab kit are not recommended here...
 - ii. ...unless you happened to buy really nice cables (\$\$\$)
4. Calibrate the VNA
 - a. Select eCal after pressing the 'Cal' button, then follow the instructions.
 - b. Be sure to use the blue SMA cables from the emag lab, not the ones you have in your lab kit.
 - c. Remember, if you change the start, stop, or center frequencies on the VNA at any point you'll have to recalibrate, so leave those alone.
 - d. Again, if you don't remember how to calibrate a VNA, ask the TA. It's better to ask than to break a \$60,000 piece of equipment. The same is especially true for the eCal module, since PSU only has one of them.

BUILD THE ANTENNAS

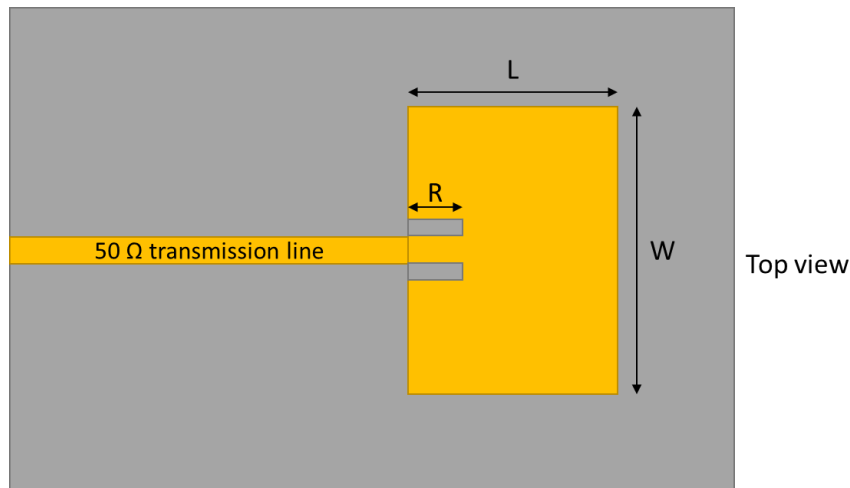


- The frequency of operation of the patch antenna of Figure 1 is determined by the length L . The center frequency will be approximately given by $f_c \approx \frac{c}{2L\sqrt{\epsilon_r}} = \frac{1}{2L\sqrt{\epsilon_r\epsilon_0\mu_0}}$
 - The above equation says that the microstrip antenna should have a length equal to one half of a wavelength *within* the dielectric medium. You'll need to calculate the wavelength of the center frequency of your antenna inside of FR4. Since we don't know exactly what the dielectric constant of FR4 is, use $\epsilon_r = 4.0, 4.1, 4.2, 4.3, \text{ and } 4.4$, then discuss how much of a difference this makes to the dimension L .
 - WiFi's carrier frequency is 2.400 GHz, and it has channels out to 2.487 GHz, so the center frequency (f_c) of the patch antenna should be right in the middle of that band.
$$f_c = \sqrt{f_1 f_2} \approx 2.443 \text{ GHz}$$
- The width W of the microstrip antenna controls the input impedance. Larger widths also can increase the bandwidth. For a square patch antenna ($L = W$) fed in the manner above, the input impedance will be on the order of 300 Ω . By increasing the width, the impedance can be reduced. However, to decrease the input impedance to 50 Ω often requires a very wide patch antenna, which takes up a lot of valuable space. The width further controls the radiation pattern.
 - It doesn't matter which direction you build the antenna (long side or short side of the board), but it will save you a lot of trouble if you align the dimension W with one of the edges of the board.
 - Don't worry about getting to 50 Ω or impedance matching yet, just make sure that W is not smaller than L .

3. Find or calculate how wide a $50\ \Omega$ transmission line is on your substrate. Most copper clad boards have a height (h) of about 1.6 mm (or about 60 mils, where 1 mil = 0.001 inch).
 - a. Connect the transmission line to the center of the patch antenna on the W-side, as shown, and run the transmission line to the edge of the board.
 - b. Solder an SMA connector's center pin to the transmission line made of copper tape, and the SMA connector's body pins to the ground plane on the other side of the board.
 - i. The top two body pins will not be soldered to anything; don't let them short-circuit to the copper tape.
 - ii. Mechanically, this is not a very robust design. The better you solder the body pins to the bottom GND plane, the less fragile this will be while handling it.
 - iii. If you happen to tear your copper tape, you can repair it with a very thin layer of solder (not a glob!)
4. Document (photograph) all antennas made by group members for the report and include all relevant design dimensions (L , W , width of transmission line)
5. Note: depending on which brand of copper tape you have, the adhesive may or may not be conductive. If you need to repair any tears or "un-cut" a section, you can add more copper tape over the top, but you may need to lightly solder the pieces of tape together.

INITIAL MEASUREMENTS AND IMPEDANCE MATCHING

1. You should have at least two completed patch antennas per group by this step.
2. If you've followed the VNA set-up instructions, two antennas can be impedance matched at the same time, both while connected to the VNA.
3. Connect one antenna to port 1 and the other to port 2.
 - a. Determine the initial operating frequency and modify each antenna as needed; this will be the frequency with the lowest S_{11} or S_{22} magnitude (the antenna connected to port 1 will use the S_{11} plots, and the antenna connected to port 2 will use the S_{22} plots).
 - i. You will need to reduce L if $f_{\text{antenna}} < f_c$, or increase L if $f_{\text{antenna}} > f_c$
 - b. Ignore S_{21} and S_{12} plots for now but keep the plots on the screen.
4. Once the antenna operates at the Wi-Fi band center frequency, recalibrate the VNA to 2.35-2.55 GHz.
5. The patch antenna was fed at the end by the transmission line. Since this typically yields a high input impedance, we would like to modify the feed (note: the "feed" of an antenna is where the transmission line touches the geometry of the antenna).
6. Since the current is low at the ends of a half-wave patch and increases in magnitude toward the center, the input impedance ($Z=V/I$) could be reduced if the patch was fed closer to the center. One method of doing this is by using an inset feed (a distance R from the end) as shown below:



7. Since the current has a sinusoidal distribution, moving the feed inwards a distance R will increase the current by $\cos\left(\frac{\pi R}{L}\right)$
8. The voltage also decreases in magnitude by the same amount that the current increases, so using Ohm's Law, the impedance scales as $Z_{IN_R} = Z_{IN_0} \cos^2\left(\frac{\pi R}{L}\right)$, where $Z_{IN_R} = 50\Omega$ and Z_{IN_0} is the measured impedance of your patch antenna.
 - a. At the antenna's operating frequency, measure the input impedance with the VNA.
 - i. Smith Chart will give you impedance and Log-Mag will give you return loss in decibels.
 - b. Calculate the length of R so that it matches your patch antenna to 50Ω . Save this number for comparison later.
 - c. Using a lot of care and precision, cut about half of R out of both sides of the patch antenna. The width of the cuts should be narrow but not so narrow that you can't see them easily.
 - i. Symmetry is important here.
 - ii. Do not alter the width of your transmission line while cutting!
 - iii. You should do this while the antenna is still connected to the VNA.
 - d. Examine the impedance match on the VNA in Log-Mag mode; continue cutting as needed to get a good impedance match.
 - i. You should easily be able to reach -20 dB return loss at or near your center frequency, and you should ideally have at least -9.5 dB return loss across the entire WiFi band (2.400 to 2.487 GHz)
 - ii. Place markers at these three points so the return loss at these points show up onscreen.
 - iii. More negative return loss (in decibels) is a better impedance match. Any student who exceeds -30 dB at their center frequency will receive 5 extra points on their final exam, provided they write their name in Sharpie on their antenna, takes one photograph of the antenna, the entire port cable, and the VNA screen showing the return loss (everything must be legible in the photo) and emails that photo to the professor (not the TA) by the time this lab's report is due. These instructions will not be repeated or located elsewhere.

MORE MEASUREMENTS AND POLARIZATION OBSERVATIONS

1. Change the S_{11} and S_{22} Smith plots to S_{12} and S_{21} LogMag plots, respectively, then align your partner's antenna with yours (facing towards each other). The S_{21} and S_{12} plots should be somewhat similar.
 - a. Separate the two antennas some fixed distance apart (the length of the VNA cables will be a limiting factor). When the antennas are directly facing each other, your S_{21} and S_{12} numbers will be at a maximum. Write this number down – it is your 0 dB baseline.
 - b. Rotate one antenna (not both) 15° out of alignment. How much did S_{21}/S_{12} drop from the 0 dB baseline? Write this down, too – the actual value of S_{21}/S_{12} is less important.
 - c. Continue rotating in 15° increments until you've rotated 90° , writing down the difference from the 0 dB baseline measurement each time.
 - d. Create an azimuth plot of the radiation pattern of your patch antennas link. It should be in polar format (use MS Excel), and you should use, at a minimum, the 6 data points gathered above, but more is better.
2. Patch antennas are linearly polarized in the far-field. Align the two patch antennas but rotate one so that it is still directly facing the other, tipped on its side. What is the S_{21}/S_{12} now? Is it lower if you face the antennas away from each other? Is this expected?
 - a. If the S_{21}/S_{12} is not as low when one antenna is on its side as it is when they are entirely facing away from each other, explain why.
 - b. Is the antenna spacing within the near-field, far-field, or somewhere in between?

DELIVERABLES FOR LAB REPORT

1. Screen captures of VNA with both antennas connected, showing S_{11} and S_{22} both below -20 dB at the design frequency.
2. Include in your report your calculation of dimensions L and R , and document dimension W (you picked it).
 - a. Photo(s) of both antennas. Write dimensions, L , W , R , as well as the transmission line length and width on the antenna with a Sharpie.
 - b. Note: this image is only an example, it doesn't represent this lab's design for a WiFi-band patch antenna nor all the required dimensions.
3. Azimuth plot (in polar format) with no less than 6 data points across 180° of rotation. You may assume the radiation pattern is symmetrical, but it's recommended that you take more measurements.
 - a. *"If you want to know exactly how long something is, only measure it once."* – R. Campbell
4. Format and organize the lab report to include all group members' names, images, plots, and calculations.
 - a. Formatting requirements to be set by lab TA.

